TITLE PAGE

- **Problem Statement ID – SIH-26004**
- **Problem Statement Title- AI-Assisted Early Detection System for Osteoarthritis (OA) Risk Markers in North Eastern Region (NER)**
- **Theme- MedTech, Biotech, HealthTech**
- **PS Category: Hardware**
- **Team ID- SIH 2026-260**
- **Team Name: DevelopMENTALS**
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CREPISENSE

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❖ Proposed Solution:

AI-Powered Wearable for Early Osteoarthritis Screening & Monitoring

Detailed Explanation

- Knee band with IMU/motion sensors monitors movement, gait, and posture.
- AI integrates sensor data with mobile app inputs (pain, stiffness) for OA risk assessment.
- Provides real-time guidance and generates digital screening reports.
- Offline data storage with healthcare dashboard sync.

Addresses the Problem

- Enables early screening in rural areas without immediate X-ray access.
- Enables objective knee movement and mobility assessment during screening and follow-up visits.
- Helps healthcare workers identify high-risk individuals for clinical evaluation.
- Lightweight, affordable, discreet, and multilingual design for underserved communities.

Innovation & Uniqueness

- Combines smart wearable, AI, and community screening in one platform.
- Focuses on personal baseline tracking and real-time preventive feedback.
- Offline-first architecture for remote deployment, integrating subjective and objective data.
- Serves as a screening/monitoring layer, complementing clinical diagnosis.

CREPISENSE identifies individuals who may require clinical evaluation without making imaging a prerequisite for initial screening.

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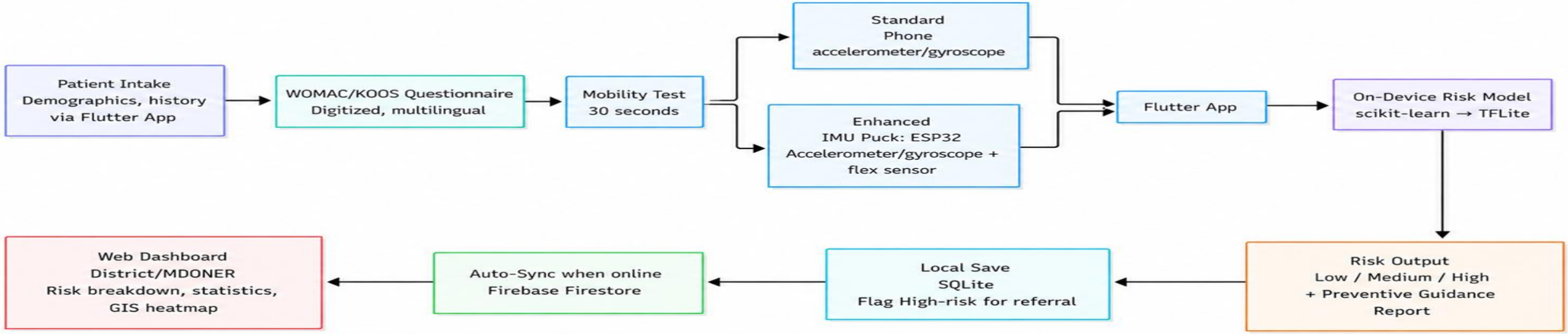
IMPLEMENTATION WORKFLOW

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SOFTWARE STACK	
COMPONENT	TECHNOLOGY / TOOL
Mobile App	Flutter (Dart)
Screening Instrument	WOMAC/KOOS (digitized)
AI Risk Model	scikit-learn → TensorFlow Lite
Local Storage	SQLite (sqflite)
Cloud Sync	Firebase Firestore + Auth
Dashboard	React + Chart.js
Localization	flutter_localizations

HARDWARE COMPONENTS	
COMPONENT	ROLE / PURPOSE
IMU Sensor (Gyroscope)	Captures knee movement and gait data
Flex / Bend Sensor	Measures knee bending angle
Microcontroller (ESP32)	Processes sensor data & manages device
Connectivity (BLE)	Wireless communication with mobile app
Vibration Motor	Provides haptic feedback & alerts
Power Source (Battery)	Powers the device (USB-C charging)
Status Indicator (LED)	Shows device status (power, connection)



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FEASIBILITY AND VIABILITY

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Feasibility Analysis

Technical: proven IMU + BLE + MCU parts

Economic: low-cost BOM, scalable build

Operational: offline-first, easy to wear

Regulatory: screening tool, not diagnostic

Challenges & Risks

Sensor noise in real-world movement

Battery life for all-day wear

User adoption & wearing compliance

Health-data privacy & security

Clinical validation & trust

Strategies to Overcome

Sensor fusion + calibration algorithms

Low-power design, efficient BLE sync

Comfortable fabric + simple app UX

Encrypt data, minimal-PII by design

Co-validate with clinicians, phased MVP

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IMPACT AND BENEFITS

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BENEFITS (Direct Advantages & Advancements)

1 ACCESSIBLE EARLY SCREENING

PROACTIVE HEALTHCARE ACCESS: Mobile screening units with cloud connectivity for initial OA detection in remote areas.

2 AI-POWERED SCREENING & RISK ASSESSMENT

AI-powered risk assessment to identify individuals who may require clinical evaluation.

3 LONGITUDINAL KNEE-FUNCTION MONITORING

Tracks changes in knee movement, gait and mobility across repeated assessments

8 PERSONALIZED GUIDANCE & EXERCISES

CUSTOMIZED CARE PLANS: Data-driven exercise guidance and gait correction tailored to individual kinematics.

IMPACTS (System & Socio-Economic Effects)

4 SUBSTANTIAL HEALTHCARE COST SAVINGS

REGION-WIDE ECONOMIC OPTIMIZATION: Drastically reduced overall system costs through early care and efficiency.

5 SIGNIFICANTLY REDUCED TRAVEL

IMPROVED QUALITY OF LIFE: Minimizing patient travel burdens, fuel use, and associated costs.

6 IMPROVED PROVIDER EFFICIENCY

SCALED CARE DELIVERY: Allowing fewer specialists to manage thousands of patients efficiently.

7 SOCIO-ECONOMIC EMPOWERMENT

INCREASED WORKFORCE PARTICIPATION: Enhanced mobility for vulnerable populations, leading to wider socio-economic impact.

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RESEARCH AND REFERENCES

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***Knee osteoarthritis screening using multimodal gait signals...* (2026)**

Result : Proves deep learning models analyzing dynamic IMU gait signals achieve **97.87% accuracy** in screening knee osteoarthritis without radiologic imaging.

Limitations:
• High image-transformation computing load
• Off-device GPU reliance (not TinyML edge native)

Ref 1

***Wearable IoT System for Monitoring Knee Angle & Plantar Force...* (2026)**

Result: Establishes clinical baselines: normal knee flexion yields **53°–72°**, while restricted gait ranges of **20°–30°** indicate early-to-moderate KOA.

Limitations:
• Wi-Fi dependent (no offline storage)
• Single-axis pressure tracking only

***A Wearable System for Knee Osteoarthritis...* (2025)**

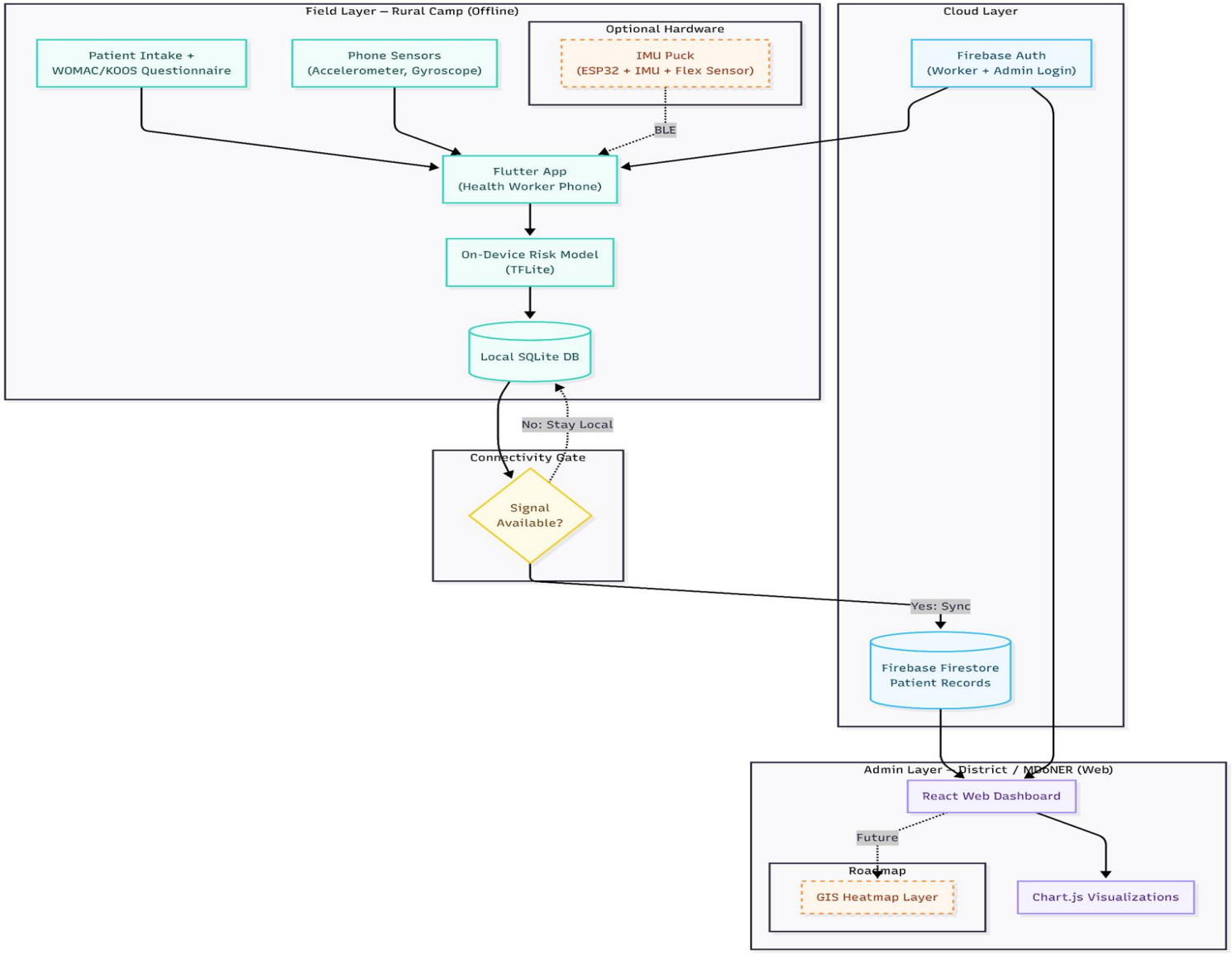
Result: Validates that combining IMUs with pressure/EMG sensors keeps functional gait feature extraction errors **below 8%** in real-world scenarios.

Limitations:
• High sensor calibration drift over time
• Costly multi-sensor hardware requirements

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ARCHITECTURE DIAGRAM

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BUSINESS MODEL CANVAS

CREPISENSE — AI-Assisted OA Risk Screening (PS 26004)

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KEY PARTNERS

- MDoNER & state NE health depts
- ASHA / ANM worker networks
- PHCs & district hospitals (referral)
- Clinicians for model validation
- NIT/BIS device certification bodies
- NGOs & CSR health programs in NER

KEY ACTIVITIES

- App + on-device TFLite model dev
- WOMAC/KOOS digitization
- Health-worker training & onboarding
- Clinical co-validation & calibration

KEY RESOURCES

- Flutter app + offline TFLite model
- IMU knee band (ESP32+BLE)
- SQLite + Firebase backend
- Trained ASHA/ANM field network

VALUE PROPOSITIONS

- Early OA risk screening with zero X-ray access needed
- "Screening before diagnosis, not grading after it"
- Offline-first — works with zero connectivity in the field
- Low-cost triage layer cuts unnecessary patient travel
- Data-driven referral prioritization for high-risk cases
- Multilingual, icon-assisted UX for field usability

CUSTOMER RELATIONSHIPS

- ASHA/ANM-assisted, in-person screening
- Dashboard support for district officers
- Ongoing app & model update cycles

CHANNELS

- Rural health camps & mobile screening units
- PHC / district hospital rollout
- MDoNER–state health dept deployment

CUSTOMER SEGMENTS

- Rural/remote NER residents at OA risk
- ASHA/ANM frontline health workers
- District health officers & MDoNER
- PHCs & hospitals receiving referrals

COST STRUCTURE

- IMU puck hardware BOM (ESP32, sensors, battery)
- App/cloud development & Firebase sync costs
- ASHA/ANM training & field rollout
- Clinical validation studies

REVENUE STREAMS

- MDoNER/govt procurement & public-health grants
- PHC/district hospital licensing of dashboard
- NGO & CSR partnership funding
- SaaS-style analytics licensing to state health depts